

time. The slave card's rendered data is sent to the master, which adds its own data and outputs the complete frame. As expected, this mode earns lower scores in 3DMark for geometry.

The only drawback of nVidia's SLI technology is that both cards have to be identical models to work. This means that if you have an older card, you cannot just buy a 7800 GTX and connect them both in SLI mode. Instead, you have to buy two 7800s and junk the existing card!

Let's take a look at ATI's multi-GPU offering.

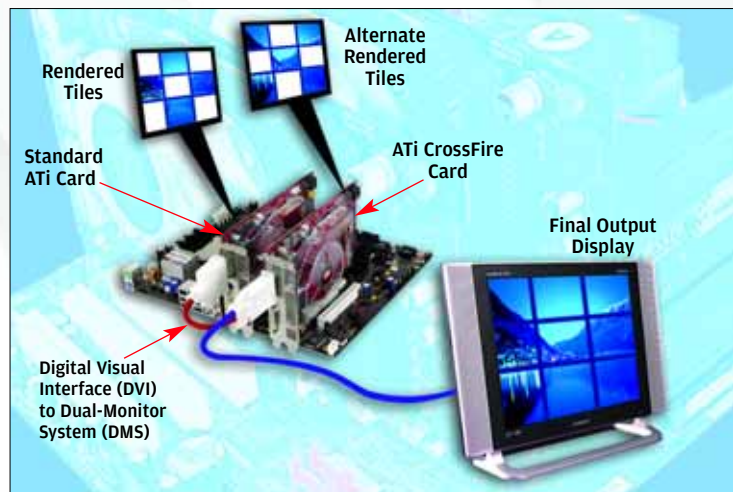
CrossFire

After a long wait, ATI finally came up with a reply to nVidia's SLI technology. Christened CrossFire, ATI's multi-GPU solution won a lot of accolades from end users because of the flexibility of being able to add a brand new, current-generation ATI CrossFire card to a system with an older PCIe card, and have them work in tandem.

Remember, in the graphics card business, six months makes your card a relic, and the latest cards always cost in excess of Rs 30,000. Considering this, a user who just spent Rs 30,000+ less than six months ago on an ATI card did not have to junk his or her card and get two new PCIe multi-GPU cards like their nVidia counterparts.

Like SLI, the CrossFire technology also has different ways in which the two cards work together.

Supertiling: Here's where the major difference between SLI and CrossFire lies. The supertiling rendering method is supported only in Direct3D rendering (Direct3D is Microsoft's proprietary Application Programming Interface (API), which is used to program many current generation games).



A visual description of how ATI's CrossFire uses supertiling across the two cards

Basically, what happens in supertiling is that each frame is broken up into 32 x 32 pixel squares. Each graphics card renders every alternate square, one by one, until the frame is completely rendered. This sharing of the rendering of square 'tiles' that make up a screen is what gave the technique its name. This method is good for equally fast graphics cards, but an older generation card such as the X800 could slow down a newer and faster CrossFire card. So, the bottomline is that both cards will run at the speed of the slower card.

Scissoring: Here the frame is divided exactly in half, and one card renders the top half while the other renders the bottom half. Unfortunately, this isn't as efficient, theoretically, as SLI's Split Frame Rendering. This is because a frame generally contains some parts with more geometry and shading, and others with much less. You might therefore end up with the card running as master finishing up its render process and then doing nothing while waiting for the other card to finish its computing!

Alternate Frame Rendering: This is exactly the same as described for nVidia's SLI technology. The only thing to remember is that since alternate frames are rendered, the total graphics sub-system speed will depend solely on the slower card.

CrossFire cards are connected via an external dangle. This connector itself introduces some drawbacks into the system. For starters, the maximum resolution for 60 Hz monitor display is 1600 x 1200 pixels. This is ridiculous considering that hardcore gamers (the people who would spend the money to buy these cards) generally use high-end monitors capable of displaying 2048 x 1536 at 70 Hz or better.



ATI's CrossFire solution is their answer to nVidia's SLI technology

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this festive season??



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